

6. Priority Scheduling (Preemptive)

```
#include <stdio.h>
#define MAX 9999
int main()
{
    int n, bt[20], pr[20], rem[20], wt[20], tat[20], ct[20];
    printf("Enter number of processes: "); scanf("%d", &n);
    for (int i=0;i<n;i++){ printf("BT and Priority of P%d: ", i+1);
scanf("%d %d", &bt[i], &pr[i]); rem[i]=bt[i]; }
    int completed=0, t=0;
    while (completed<n){
        int idx=-1;
        for (int i=0;i<n;i++) if(rem[i]>0 && (idx==-1 ||
pr[i]<pr[idx])) idx=i;
        rem[idx]--; t++;
        if (rem[idx]==0){ ct[idx]=t; tat[idx]=ct[idx]-0;
wt[idx]=tat[idx]-bt[idx]; completed++; }
    }
    printf("P\tBT\tPR\tWT\tTAT\n");
    for (int i=0;i<n;i++) printf("P%d\t%d\t%d\t%d\t%d\n", i+1,
bt[i], pr[i], wt[i], tat[i]);
    return 0;
}
```

Sample Input: 3 / 5 2 / 3 1 / 8 3

Output:

```
Enter number of processes: BT and Priority of P1: BT and Priority of
P2: BT and Priority of P3: P      BT      PR      WT      TAT
P1   5      2      3      8
P2   3      1      0      3
P3   8      3      8     16
```